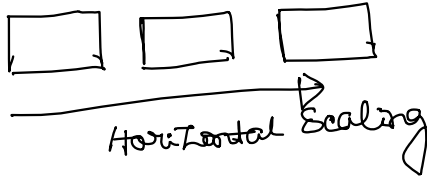


# System Design

## 1. Horizontal Vs Vertical Scaling

↳ Scale out

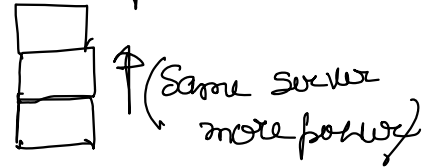
↳ Adding more servers



Scale up

→ Adding more power to servers

eg → CPU RAMs



 Vertical

Traffic low

Simplicity

Horizontal

For large scale app

Huge Traffic

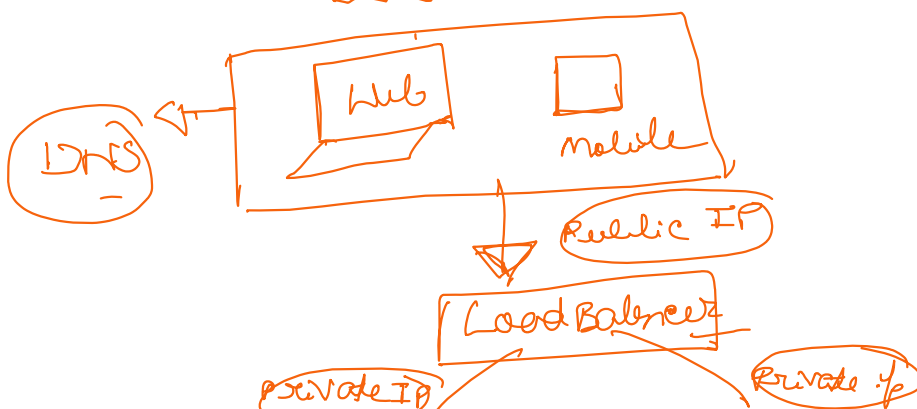
## Limitation of Vertical Scaling

→ Impossible to add unlimited CPU & RAM to a single server

→ No Backup (failover)

→ If one server goes down, The website/server completely goes down.

## Load Balancer:-



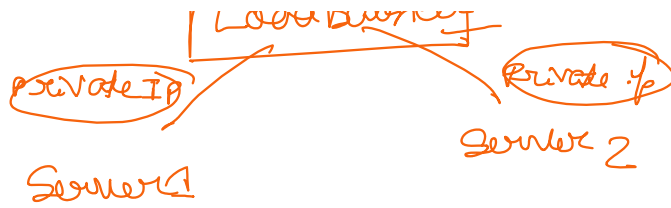
## → Pros design

↳ Users connected directly

→ When server is offline

→ Also when many users access at same time

→ Slow resp  
→ Fail to connect



→ Fail to connect

→ Load Balancer communicates with web server via private ip

Load Balancer + 2nd web server

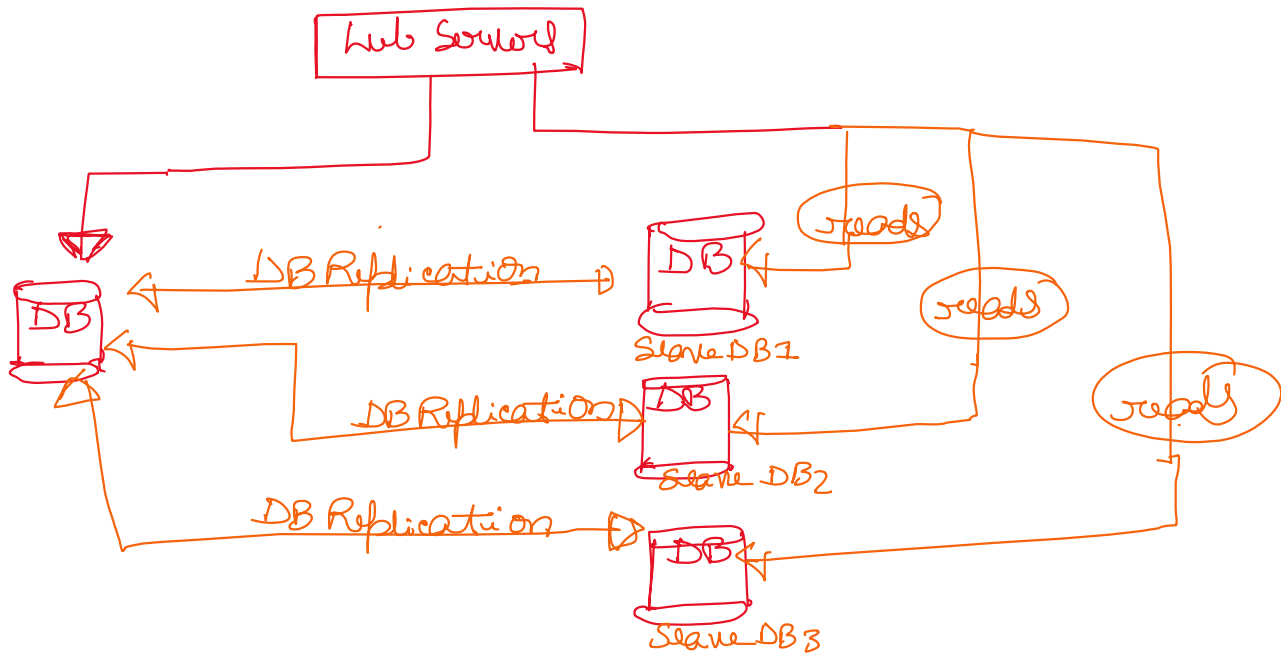
(11) What if Data Base down?

DB Replication

- master - original DB
- Slave → copy

Write ops - insert, delete, update

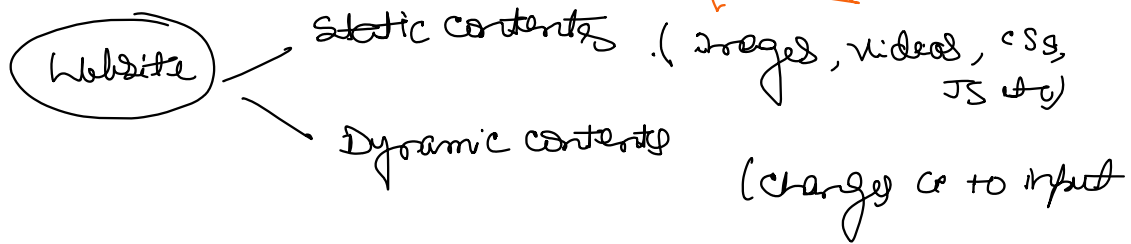
Read ops → get copies from master



Advantages of DB Replication

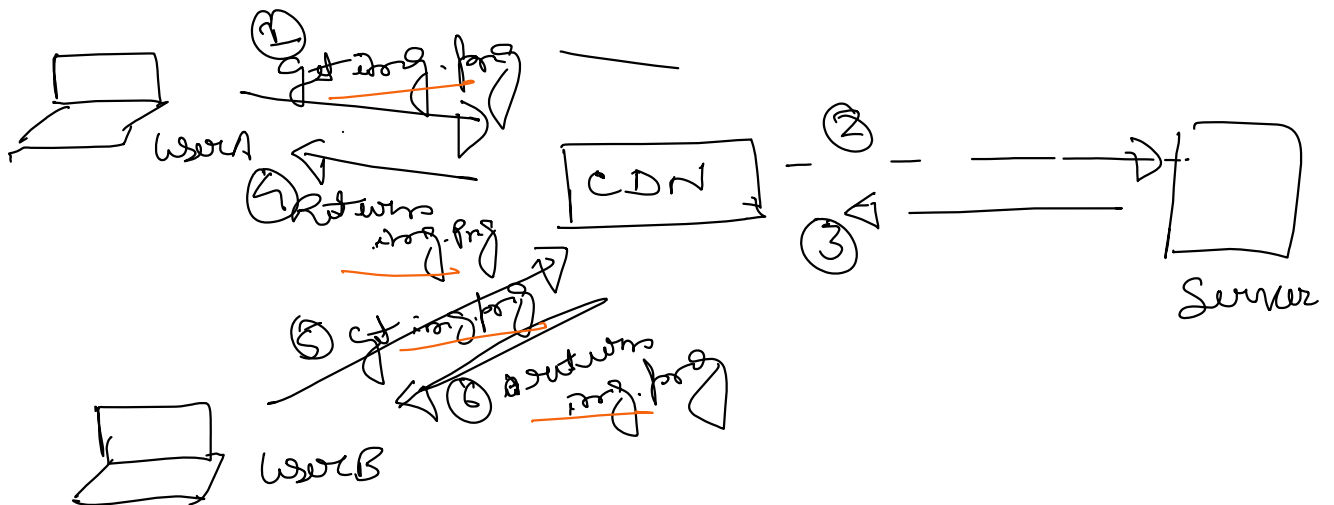
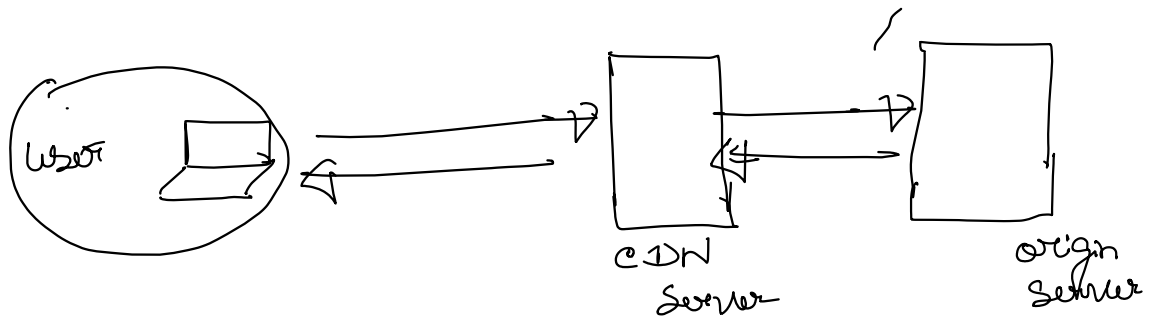
- Better performance
- Reliability
- High Availability

## VII CDN → (Content delivery Network)



### How CDN Works?

CDN (closest to user) Static content



### III Things to consider for using CDN

- cost — Get rid of infrequent used contents

- Cost — Get rid of infrequent used contents
- Expiry time — not too low, not too high
- CDN Fallback/Backup — Back up when CDN fails
- Invalidating files — API (CDN vendor)  
↳ versioning

Vid-10

### Stateless and Stateful Architecture

• Session data / State:-

Data that represents user's session on hub

- Identify a user  
- Authenticate a user

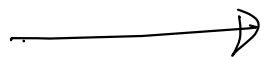
User-A



Hub Server (1)

State of User 1

User-B



Hub Server (2)

State of info of User-2

Stateless hub

→ Does not remember client data

- Does not store any info

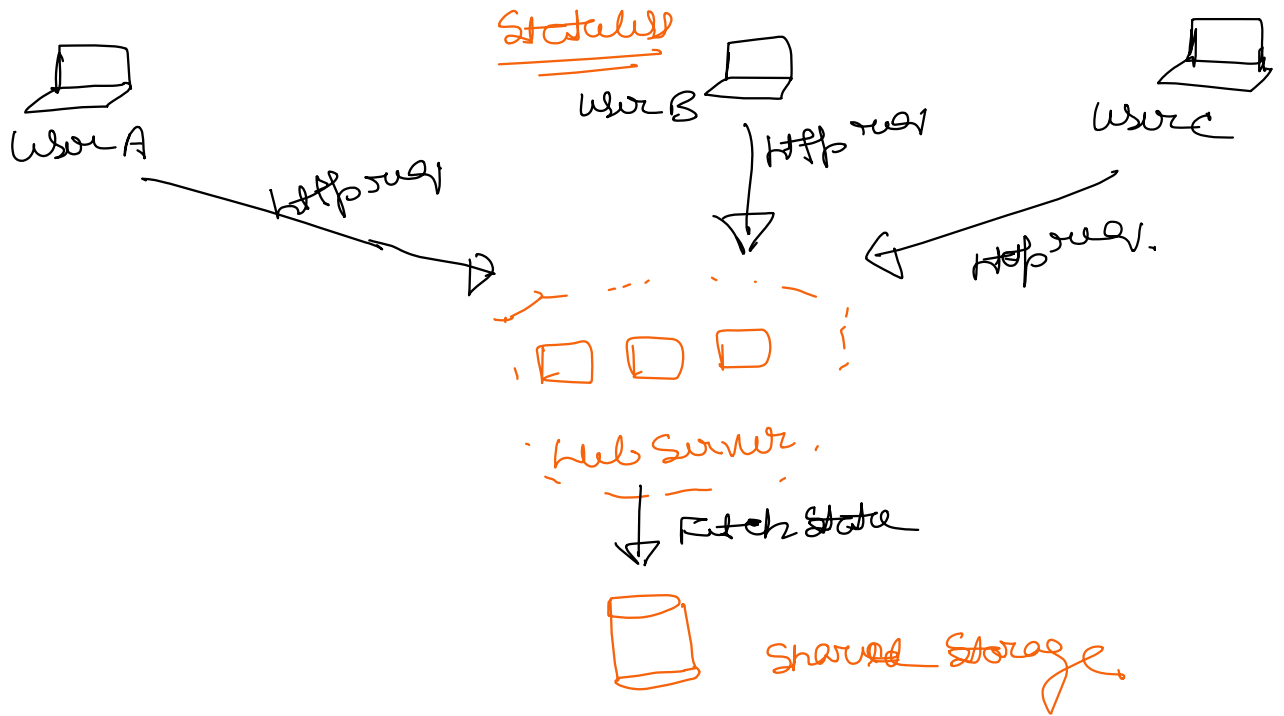
Stateful hub

- Remembers

→ Stores

- Overhead, not simple

- Simple, robust  
 - overhead, not simple



Now our service looks like

- multiple server
- Load Balancer
- master slave DB
- cache DB
- CDN
- distributed stateful web